

BA Capstone Project: Checkpoint 1

Data Fundamentals & SQL Querying

Group Name: MANTA RAY

Course & Section: BED 106 - Business Analytics

Date: [SUBMISSION DATE]

Team Members & Roles:

- Sugar-Rey R. Garcia — Project Lead / Analyst
- Rudelyn Joy Abanador— Data Engineer
- Joel Madula — Statistician / Modeler

Task 1.1 — Business Problem Statement

Business Domain & Problem Investigation

This project belongs to the Retail & Sales Analytics domain. The business problem being investigated is Sales Performance & Revenue Forecasting — specifically, identifying which products and regions generate the most revenue, analyzing sales trends over time, and forecasting future revenue to support business decision-making. Using historical transactional data from a retail superstore, this project aims to uncover patterns in sales, profit, and discount behavior across different product categories and geographic regions.

Business Impact & Significance

Understanding sales performance is critical for any retail business. Without clear visibility into which products, regions, and time periods drive the most revenue, businesses risk over-investing in low-performing areas and missing growth opportunities. Accurate revenue forecasting enables management to plan inventory, allocate budgets, and set realistic sales targets. This project directly addresses the need for data-driven decision-making in retail operations, which can lead to improved profitability, reduced waste from overstocking, and more targeted marketing efforts toward high-value customer segments.

Key Business Questions

- 1. Which product categories and regions generate the highest total sales and profit?
- 2. Does offering higher discounts lead to lower profit margins?
- 3. What are the monthly and seasonal sales trends over time?
- 4. Which customer segments contribute the most to total revenue?
- 5. What will next quarter's sales look like based on historical trends?

Data Source Overview

The dataset planned for this project is the Sample Superstore Sales Dataset, a publicly available retail transaction dataset sourced from Kaggle (www.kaggle.com). It contains structured, tabular data consisting of 9,994 rows of order-level records with 21 columns covering customer information, geographic data, product categories, and financial metrics such as sales, profit, discount, and quantity. The data represents multiple years of retail transactions, making it suitable for time-series trend analysis and regression-based forecasting.

Task 1.2 — Dataset Documentation

Data Source Report

Attribute	Details
Dataset Name	Sample Superstore Sales Dataset
Source URL/File	https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
Date Accessed	19/07/2026
License Type	CC0: Public Domain (No restrictions on use)

Data Dictionary

Column Name	Data Type	Description
Order ID	VARCHAR(20)	Unique identifier for each order transaction
Order Date	DATE	The date when the order was placed
Ship Date	DATE	The date when the order was shipped
Ship Mode	VARCHAR(30)	Shipping method selected (e.g., First Class, Standard Class)
Customer ID	VARCHAR(20)	Unique identifier for each customer
Customer Name	VARCHAR(100)	Full name of the customer
Segment	VARCHAR(30)	Customer market segment (Consumer, Corporate, Home Office)
Country	VARCHAR(50)	Country where the order was delivered
City	VARCHAR(50)	City where the order was delivered
State	VARCHAR(50)	State where the order was delivered
Region	VARCHAR(20)	Geographic region (East, West, Central, South)
Product ID	VARCHAR(20)	Unique identifier for each product

Column Name	Data Type	Description
Category	VARCHAR(30)	Main product category (Furniture, Office Supplies, Technology)
Sub-Category	VARCHAR(30)	Product sub-category (e.g., Chairs, Phones, Binders)
Product Name	VARCHAR(200)	Full descriptive name of the product
Sales	DECIMAL(10,2)	Total sales revenue for the order line item
Quantity	INT	Number of units ordered
Discount	DECIMAL(5,2)	Discount rate applied to the order (0.0 to 1.0)
Profit	DECIMAL(10,2)	Net profit earned from the order line item

Preliminary Data Quality Assessment

- Missing Values:** *The Sample Superstore dataset has no missing values across all 9,994 rows and 21 columns. All fields are fully populated with no null or empty entries detected.*
- Duplicates:** *No exact duplicate rows were found. However, the same Order ID may appear multiple times when a single order contains multiple products, which is expected and normal behavior in order-line transactional datasets.*
- Inconsistencies:** *Dates are stored in MM/DD/YYYY format requiring conversion to YYYY-MM-DD for MySQL. Postal Code contains numeric values stored as text which must be handled as VARCHAR. A number of records show negative Profit*

values indicating loss-making orders — these are valid data points and are retained for analysis.

- **Cleaning Steps Applied:** Date columns (Order Date, Ship Date) were reformatted to YYYY-MM-DD for MySQL compatibility. Postal Code was cast as VARCHAR. Column names were standardized by removing spaces and special characters. The flat dataset was split into three relational tables: customers, products, and orders, with primary keys and foreign keys assigned accordingly.

Raw Dataset Preview

The first 10–20 rows of the raw Superstore CSV file

Screenshot 1 – <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

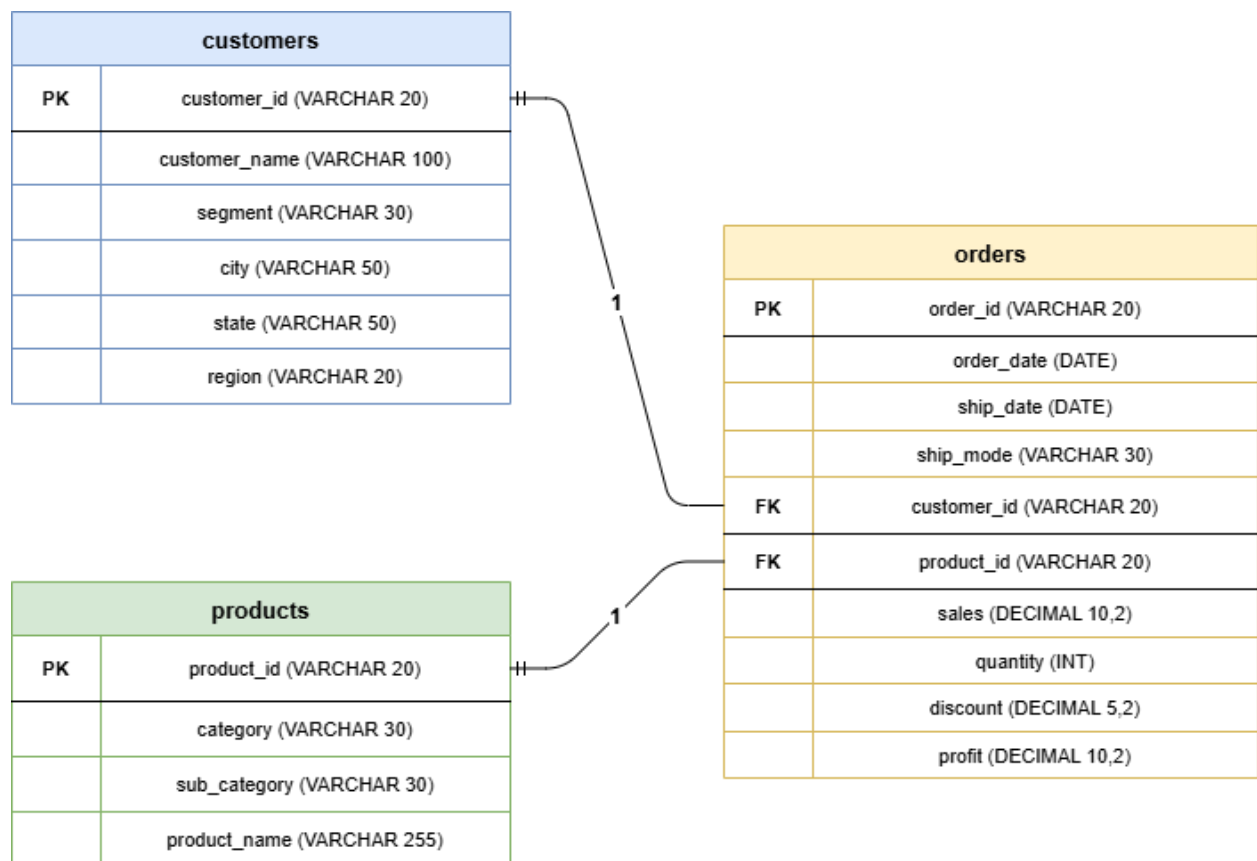
Sample - Superstore.csv (2.29 MB)																				21 of 21 columns	
Detail		Compact																		Column	
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount		
1	CA-2016-100736	11/8/2016	11/11/2016	Second Class	CS-10029	Claire Soto	Consumer	United States	Henderson	Kentucky	40428	South	POB-SO-10007760	Furniture	Bookcases	Book Shelves Collection Bookcase	261.96	3	0		
2	CA-2016-100736	11/8/2016	11/11/2016	Second Class	CS-10029	Claire Soto	Consumer	United States	Henderson	Kentucky	40428	South	POB-SO-10004354	Furniture	Chairs	Non-Deluxe Fabric Upholstered Stacking Chair, Padded Back	121.84	3	0		
3	CA-2016-138600	6/12/2016	6/16/2016	Second Class	CS-13860	Derron Van Hult	Corporate	United States	Los Angeles	California	90026	West	OFF-1A-10000348	Office Supplies	Labels	Self-Adhesive Address Labels for Specimens by Universal	16.62	2	0		
4	US-2015-100896	10/11/2015	10/18/2015	Standard Class	CS-08025	Sean O'Sullivan	Consumer	United States	Fort Lauderdale	Florida	33311	South	POB-1A-10000577	Furniture	Tables	Bradford Café Series Slim Rectangular Table	927.5775	5	0.45		
5	US-2015-100896	10/11/2015	10/18/2015	Standard Class	CS-08025	Sean O'Sullivan	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-SI-10000760	Office Supplies	Storage	Glass Fold In Mail Cart System	32.388	2	0.2		
6	CA-2014-110810	6/9/2014	6/14/2014	Standard Class	BS-11710	Broscoe Huffman	Consumer	United States	Los Angeles	California	90002	West	POB-FI-10001487	Furniture	Furnishings	Glass Expressions Wood and Plastic Desk Accessories Cherry Wood	48.86	7	0		
7	CA-2014-110810	6/9/2014	6/14/2014	Standard Class	BS-11710	Broscoe Huffman	Consumer	United States	Los Angeles	California	90002	West	OFF-AI-10000003	Office Supplies	Art	Newell 327	7.28	4	0		
8	CA-2014-110810	6/9/2014	6/14/2014	Standard Class	BS-11710	Broscoe Huffman	Consumer	United States	Los Angeles	California	90002	West	ICC-FI-10000276	Technology	Phones	Apple iPhone 11 Phone 1107 Phone	987.182	4	0.2		
9	CA-2014-110810	6/9/2014	6/14/2014	Standard Class	BS-11710	Broscoe Huffman	Consumer	United States	Los Angeles	California	90002	West	OFF-SI-10000370	Office Supplies	Binders	One-Angle View Binders with Locking Rings by Samsill	10.584	3	0.2		
10	CA-2014-110810	6/9/2014	6/14/2014	Standard Class	BS-11710	Broscoe Huffman	Consumer	United States	Los Angeles	California	90002	West	OFF-AP-10000000	Office Supplies	Appliances	Neila F0300011 & Outlet Surge	114.9	5	0		
11	CA-2014-110810	6/9/2014	6/14/2014	Standard Class	BS-11710	Broscoe Huffman	Consumer	United States	Los Angeles	California	90002	West	POB-1A-10001029	Furniture	Tables	Chromcraft Rectangular Conference Table	1786.184	5	0.2		
12	CA-2014-110810	6/9/2014	6/14/2014	Standard Class	BS-11710	Broscoe Huffman	Consumer	United States	Los Angeles	California	90002	West	ICC-FI-10000003	Technology	Phones	Apple iPhone 11 Conference Table by Samsill	911.424	4	0.2		
13	CA-2017-110402	4/15/2017	4/20/2017	Standard Class	AA-10600	Andrew Allen	Consumer	United States	Concord	North Carolina	28607	South	OFF-PA-10000365	Office Supplies	Power	Kerox 1907	15.502	3	0.2		
14	CA-2016-160389	12/5/2016	12/16/2016	Standard Class	JK-12670	Jesse Mathis	Consumer	United States	Seattle	Washington	98105	West	OFF-SI-10000006	Office Supplies	Binders	Yellowline Plastic Desk Binding Machine	487.876	3	0.2		
15	US-2015-100890	11/22/2015	11/26/2015	Standard Class	HF-14815	Harold Pavilion	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10000311	Office Supplies	Appliances	Neila F0300011 Outlet Surge	60.81	5	0.8		

Screenshot 2 – Downloaded CSV file (Sample – Superstore.csv)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Row ID	Order ID	Order Date	Ship Date	Ship Mod	Customer	Customer Segment	Country	City	State	Postal Co	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	
2		1	CA-2016-1	11/08/2016	11/11/2016	Second CI	CG-12520	Claire Gu	Consume	United St	Hendersc	Kentucky	42420	South	FUR-BO-1	Furniture Bookcase	Bush Som	261.96	2	0	41.9136
3		2	CA-2016-1	11/08/2016	11/11/2016	Second CI	CG-12520	Claire Gu	Consume	United St	Hendersc	Kentucky	42420	South	FUR-CH-1	Furniture Chairs	Hon Delu	731.94	3	0	219.582
4		3	CA-2016-1	06/12/2016	6/16/2016	Second CI	DV-13045	Darrin Va	Corporate	United St	Los Angel	California	90036	West	OFF-LA-10	Office Suj Labels	Self-Adhe	14.62	2	0	6.8714
5		4	US-2015-1	10/11/2015	10/18/2015	Standard	SO-20335	Sean O'Di	Consume	United St	Fort Laud	Florida	33311	South	FUR-TA-1	Furniture Tables	Bretford C	957.5775	5	0.45	-383.031
6		5	US-2015-1	10/11/2015	10/18/2015	Standard	SO-20335	Sean O'Di	Consume	United St	Fort Laud	Florida	33311	South	OFF-ST-10	Office Suj Storage	Eldon Fol	22.368	2	0.2	2.5164
7		6	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	FUR-FU-1	Furniture Furnishin	Eldon Exp	48.86	7	0	14.1694
8		7	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	OFF-AR-1	Office Suj Art	Newell 3	7.28	4	0	1.9656
9		8	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	TEC-PH-10	Technolo Phones	Mitel 532	907.152	6	0.2	90.7152
10		9	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	OFF-BI-10	Office Suj Binders	DXL Angle	18.504	3	0.2	5.7825
11		10	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	OFF-AP-1	Office Suj Appliance	Belkin F5	114.9	5	0	34.47
12		11	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	FUR-TA-1	Furniture Tables	Chromcra	1706.184	9	0.2	85.3092
13		12	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	TEC-PH-10	Technolo Phones	Konftel 2	911.424	4	0.2	68.3568
14		13	CA-2017-1	4/15/2017	4/20/2017	Standard	AA-10480	Andrew A	Consume	United St	Concord	North Car	28027	South	OFF-PA-1	Office Suj Paper	Xerox 196	15.552	3	0.2	5.4432
15		14	CA-2016-1	12/05/2016	12/10/2016	Standard	IM-15070	Irene Mai	Consume	United St	Seattle	Washingt	98103	West	OFF-BI-10	Office Suj Binders	Fellowes	407.976	3	0.2	132.5922
16		15	US-2015-1	11/22/2015	11/26/2015	Standard	HP-14815	Harold Pa	Home Off	United St	Fort Wort	Texas	76106	Central	OFF-AP-1	Office Suj Appliance	Holmes R	68.81	5	0.8	-123.858
17		16	US-2015-1	11/22/2015	11/26/2015	Standard	HP-14815	Harold Pa	Home Off	United St	Fort Wort	Texas	76106	Central	OFF-BI-10	Office Suj Binders	Storex Du	2.544	3	0.8	-3.816
18		17	CA-2014-1	11/11/2014	11/18/2014	Standard	PK-19075	Pete Kriz	Consume	United St	Madison	Wisconsi	53711	Central	OFF-ST-10	Office Suj Storage	Stur-D-Stc	665.88	6	0	13.3176
19		18	CA-2014-1	5/13/2014	5/15/2014	Second CI	AG-10270	Alejandro	Consume	United St	West Jorc	Utah	84084	West	OFF-ST-10	Office Suj Storage	Fellowes	55.5	2	0	9.99
20		19	CA-2014-1	8/27/2014	09/01/2014	Second CI	DZ-21925	Zuschuss	Consume	United St	San Franc	California	94109	West	OFF-AR-1	Office Suj Art	Newell 34	8.56	2	0	2.4824

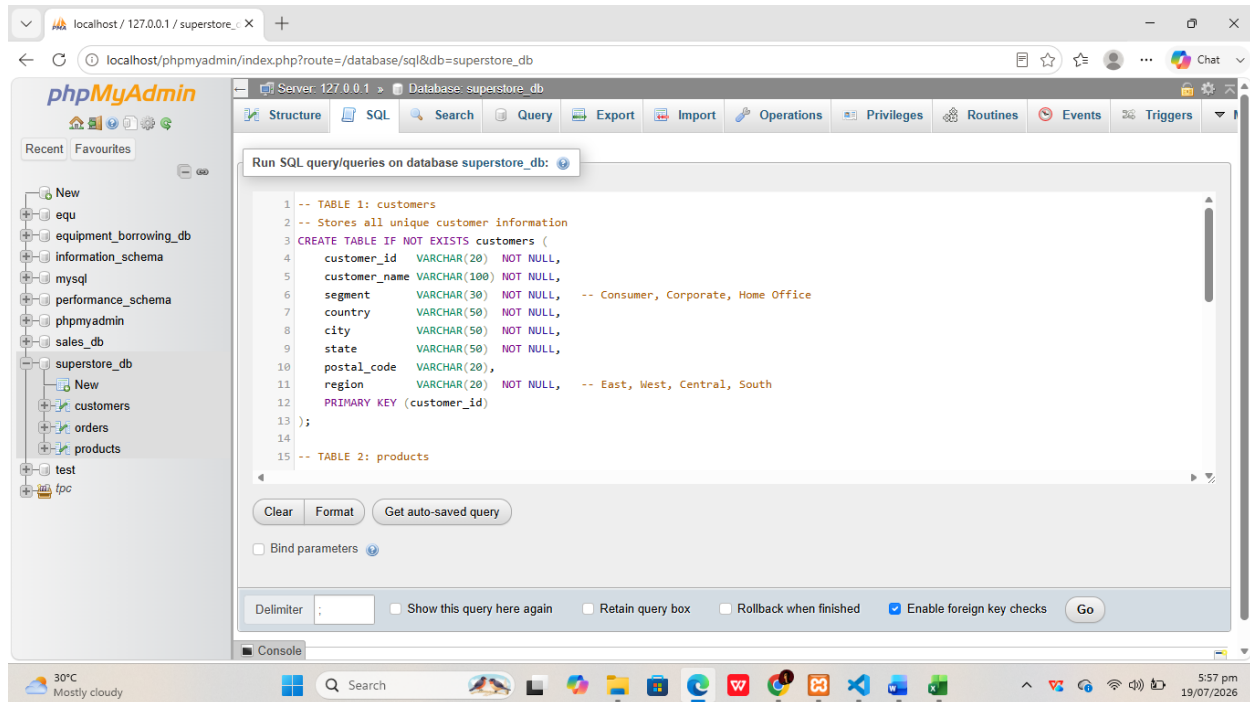
Task 1.3 — Relational Database Design

Entity-Relationship Diagram (ERD)



Database Schema & Population

Screenshot 1 — CREATE TABLE statements

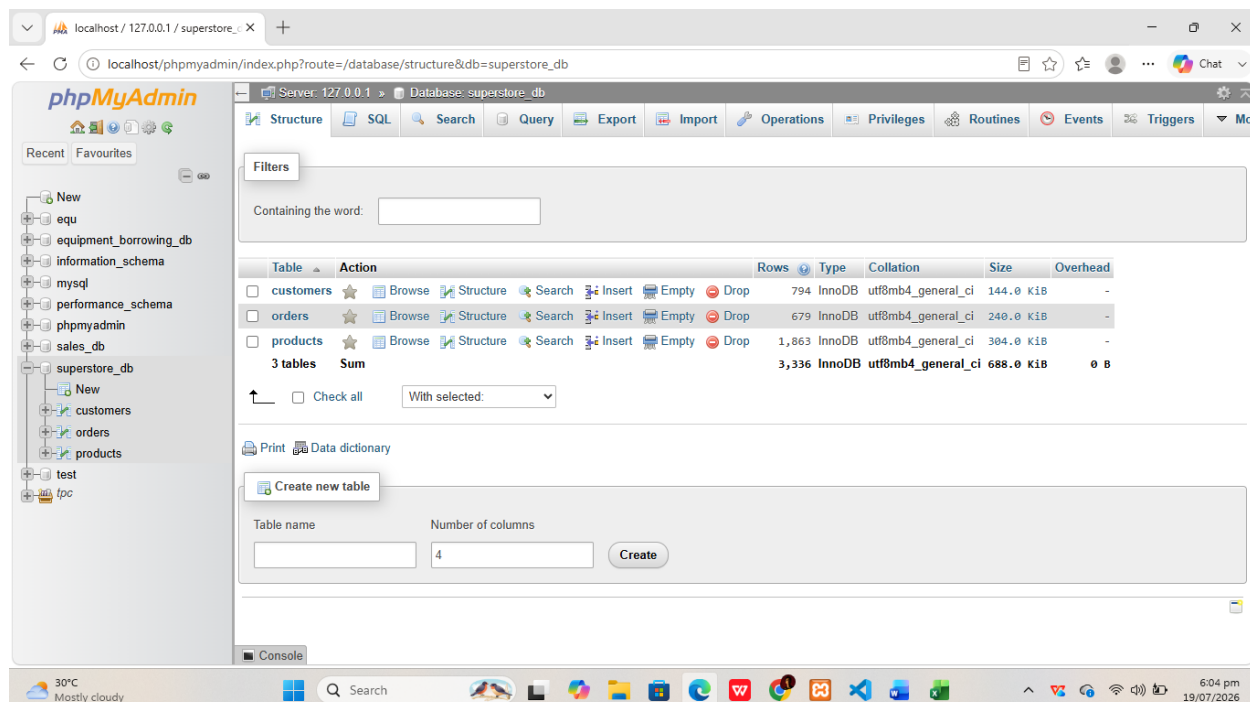


The screenshot shows the phpMyAdmin interface for the 'superstore_db' database. The 'SQL' tab is active, displaying the following SQL queries:

```
1 -- TABLE 1: customers
2 -- Stores all unique customer information
3 CREATE TABLE IF NOT EXISTS customers (
4   customer_id VARCHAR(20) NOT NULL,
5   customer_name VARCHAR(100) NOT NULL,
6   segment VARCHAR(30) NOT NULL, -- Consumer, Corporate, Home Office
7   country VARCHAR(50) NOT NULL,
8   city VARCHAR(50) NOT NULL,
9   state VARCHAR(50) NOT NULL,
10  postal_code VARCHAR(20),
11  region VARCHAR(20) NOT NULL, -- East, West, Central, South
12  PRIMARY KEY (customer_id)
13 );
14
15 -- TABLE 2: products
```

The interface includes a sidebar with a database tree, a top navigation bar with tabs like Structure, SQL, Search, Query, Export, Import, Operations, Privileges, Routines, Events, and Triggers, and a bottom status bar showing the server version (127.0.0.1) and the database name (superstore_db).

Screenshot 2 — Three tables in the schema



The screenshot shows the phpMyAdmin interface for the 'superstore_db' database, displaying the 'Structure' tab. The 'Filters' section is visible, and the table list shows three tables: customers, orders, and products. The table list includes columns for Table, Action, Rows, Type, Collation, Size, and Overhead.

Table	Action	Rows	Type	Collation	Size	Overhead
customers	Browse Structure Search Insert Empty Drop	794	InnoDB	utf8mb4_general_ci	144.0 KiB	-
orders	Browse Structure Search Insert Empty Drop	679	InnoDB	utf8mb4_general_ci	240.0 KiB	-
products	Browse Structure Search Insert Empty Drop	1,863	InnoDB	utf8mb4_general_ci	304.0 KiB	-
3 tables	Sum	3,336	InnoDB	utf8mb4_general_ci	688.0 KiB	0 B

The interface also includes a sidebar with a database tree, a top navigation bar with tabs like Structure, SQL, Search, Query, Export, Import, Operations, Privileges, Routines, Events, and Triggers, and a bottom status bar showing the server version (127.0.0.1) and the database name (superstore_db).

Screenshot 3 — Table previews

- Customers Table

The screenshot shows the phpMyAdmin interface for the 'superstore_db' database. The 'customers' table is selected, and the 'Table: customers' view is active. The interface displays a list of 25 rows (0-24) of customer data. The table structure is as follows:

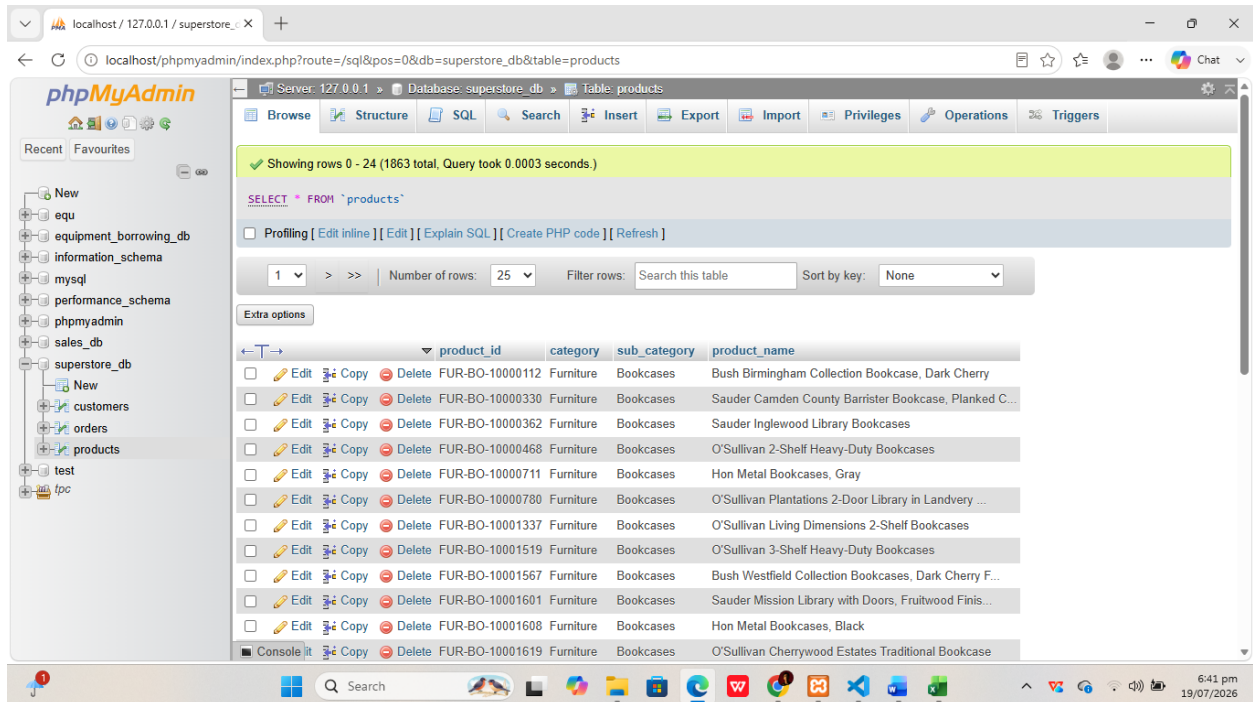
customer_id	customer_name	segment	country	city	state	postal_code	region
AA-10315	Alex Avila	Consumer	United States	Minneapolis	Minnesota	55407	Central
AA-10375	Allen Arnold	Consumer	United States	Mesa	Arizona	85204	West
AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South
AA-10645	Anna Andreadi	Consumer	United States	Chester	Pennsylvania	19013	East
AB-10015	Aaron Bergman	Consumer	United States	Seattle	Washington	98103	West
AB-10060	Adam Bellavance	Home Office	United States	New York City	New York	10009	East
AB-10105	Adrian Barton	Consumer	United States	Phoenix	Arizona	85023	West
AB-10150	Aimee Bixby	Consumer	United States	Long Beach	New York	11561	East
AB-10165	Alan Barnes	Consumer	United States	Los Angeles	California	90036	West
AB-10255	Alejandro Ballentine	Home Office	United States	Lorain	Ohio	44052	East
AB-10600	Ann Blume	Corporate	United States	Tucson	Arizona	85705	West
AC-10420	Alyssa Crouse	Corporate	United States	San Francisco	California	94122	West

- Orders Table

The screenshot shows the phpMyAdmin interface for the 'superstore_db' database. The 'orders' table is selected, and the 'Table: orders' view is active. The interface displays a list of 25 rows (0-24) of order data. The table structure is as follows:

order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
CA-2014-101476	2014-09-12	2014-09-13	First Class	SD-20485	TEC-MA-10000029	69.99	1	0.00	30.10
CA-2014-103849	2014-05-11	2014-05-16	Standard Class	PG-18895	FUR-FU-10000723	66.11	4	0.60	-84.29
CA-2014-103849	2014-05-11	2014-05-16	Standard Class	PG-18895	TEC-AC-10001465	58.11	2	0.20	7.26
CA-2014-103849	2014-05-11	2014-05-16	Standard Class	PG-18895	TEC-PH-10002597	100.79	1	0.20	6.30
CA-2014-104269	2014-03-01	2014-03-06	Second Class	DB-13060	FUR-CH-10004063	457.57	2	0.20	51.48
CA-2014-105893	2014-11-11	2014-11-18	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0.00	13.32
CA-2014-106376	2014-12-05	2014-12-10	Standard Class	BS-11590	OFF-AR-10002671	1113.02	8	0.20	111.30
CA-2014-106376	2014-12-05	2014-12-10	Standard Class	BS-11590	TEC-PH-10002726	167.97	4	0.20	62.99
CA-2014-110072	2014-10-22	2014-10-28	Standard Class	MG-17680	FUR-FU-10000521	93.89	4	0.20	12.91
CA-2014-111003	2014-06-01	2014-06-06	Standard Class	CR-12625	OFF-AR-10002135	289.20	6	0.00	83.87
CA-2014-111003	2014-06-01	2014-06-06	Standard Class	CR-12625	OFF-BI-10001072	45.48	3	0.00	20.92
CA-2014-111451	2014-12-26	2014-12-28	First Class	KL-16555	FUR-CH-10001891	230.35	3	0.20	20.16

- **Products Table**



Task 1.4 — SQL Query Report

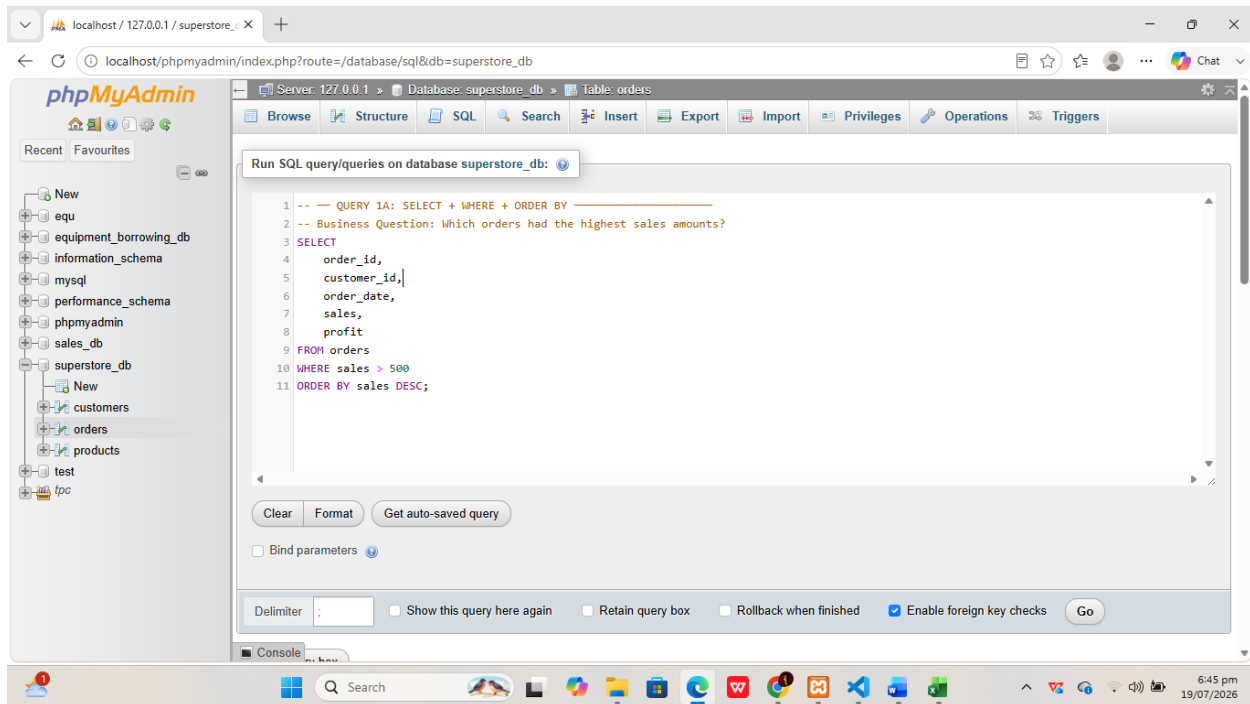
1. Basic Data Retrieval (2 Queries)

Must include *SELECT*, *WHERE*, and *ORDER BY* clauses.

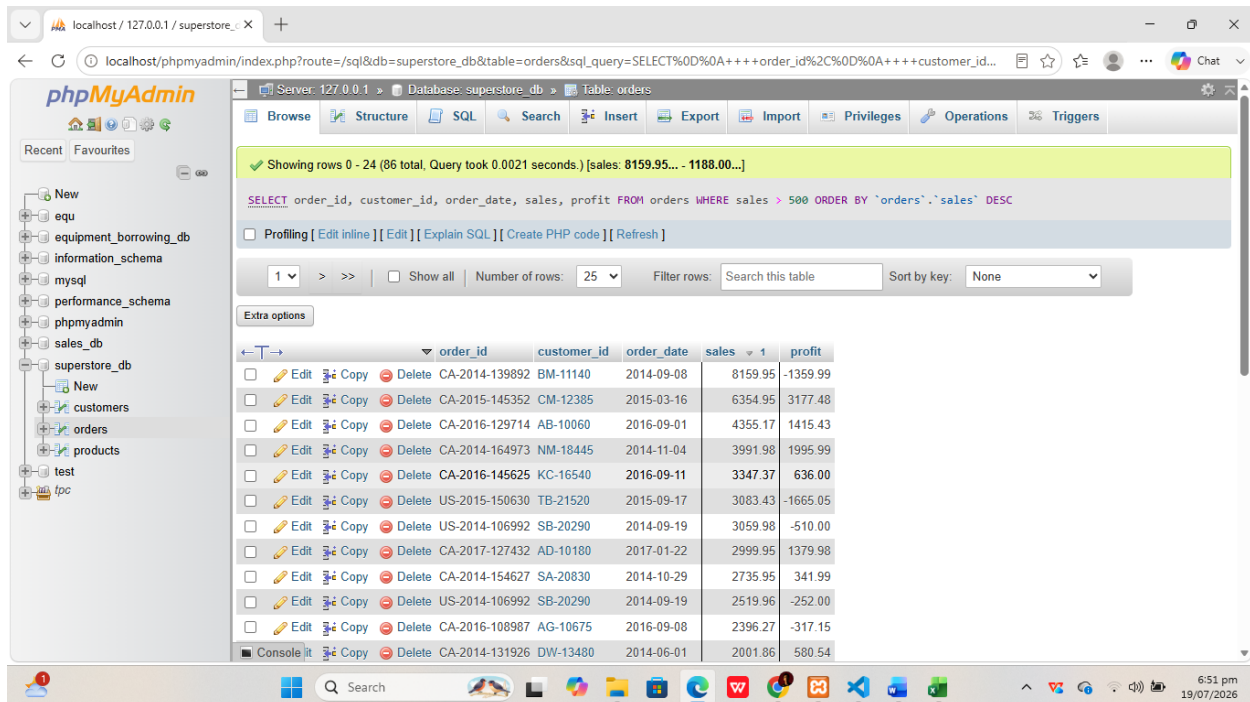
QUERY 1B: *SELECT* + *WHERE* + *ORDER BY*

- -- Business Question: What orders were placed in 2017?

- **Screenshot of SQL Code**



- **Result Screenshot**



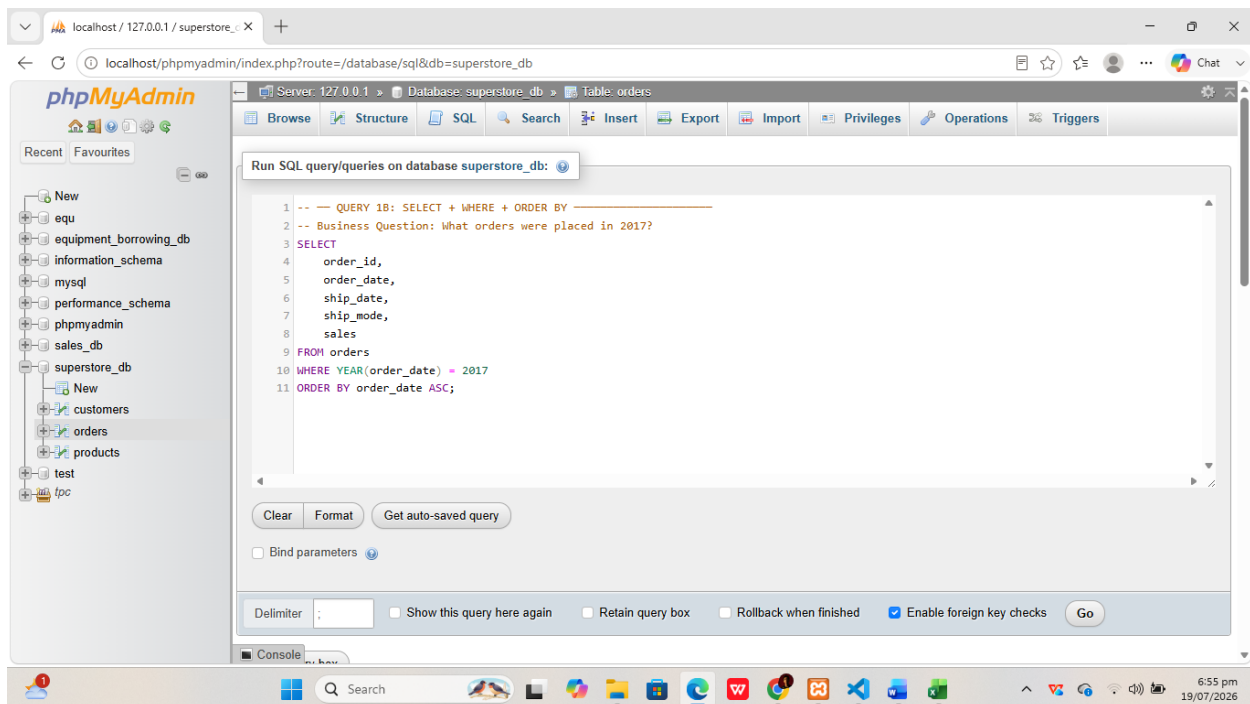
- **Business Interpretation**

Query 1A: Retrieves all orders where the sales amount exceeds 500, sorted from highest to lowest. This identifies high-value transactions that contribute most significantly to total revenue, helping management understand patterns among large purchases such as which products or regions tend to produce high-value orders.

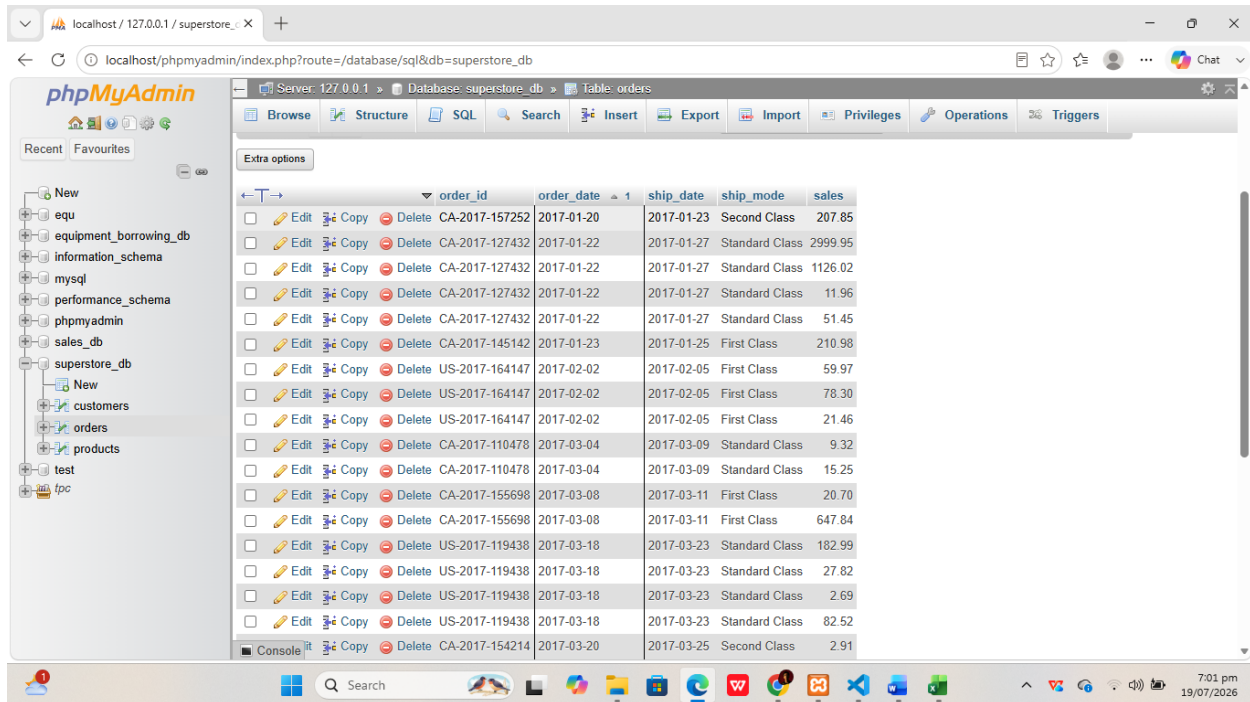
QUERY 1B: SELECT + WHERE + ORDER BY

-- Business Question: What orders were placed in 2017?

- **Screenshot of SQL Code**



- **Result Screenshot**



	order_id	order_date	ship_date	ship_mode	sales
☐	CA-2017-157252	2017-01-20	2017-01-23	Second Class	207.85
☐	CA-2017-127432	2017-01-22	2017-01-27	Standard Class	2999.95
☐	CA-2017-127432	2017-01-22	2017-01-27	Standard Class	1126.02
☐	CA-2017-127432	2017-01-22	2017-01-27	Standard Class	11.96
☐	CA-2017-127432	2017-01-22	2017-01-27	Standard Class	51.45
☐	CA-2017-145142	2017-01-23	2017-01-25	First Class	210.98
☐	US-2017-164147	2017-02-02	2017-02-05	First Class	59.97
☐	US-2017-164147	2017-02-02	2017-02-05	First Class	78.30
☐	US-2017-164147	2017-02-02	2017-02-05	First Class	21.46
☐	CA-2017-110478	2017-03-04	2017-03-09	Standard Class	9.32
☐	CA-2017-110478	2017-03-04	2017-03-09	Standard Class	15.25
☐	CA-2017-155698	2017-03-08	2017-03-11	First Class	20.70
☐	CA-2017-155698	2017-03-08	2017-03-11	First Class	647.84
☐	US-2017-119438	2017-03-18	2017-03-23	Standard Class	182.99
☐	US-2017-119438	2017-03-18	2017-03-23	Standard Class	27.82
☐	US-2017-119438	2017-03-18	2017-03-23	Standard Class	2.69
☐	US-2017-119438	2017-03-18	2017-03-23	Standard Class	82.52
☐	CA-2017-154214	2017-03-20	2017-03-25	Second Class	2.91

- **Business Interpretation:**

Query 1B: Filters all orders placed in 2017 in chronological order, allowing the business to review the full transaction history for a specific fiscal year and compare annual sales volumes to track business growth over time.

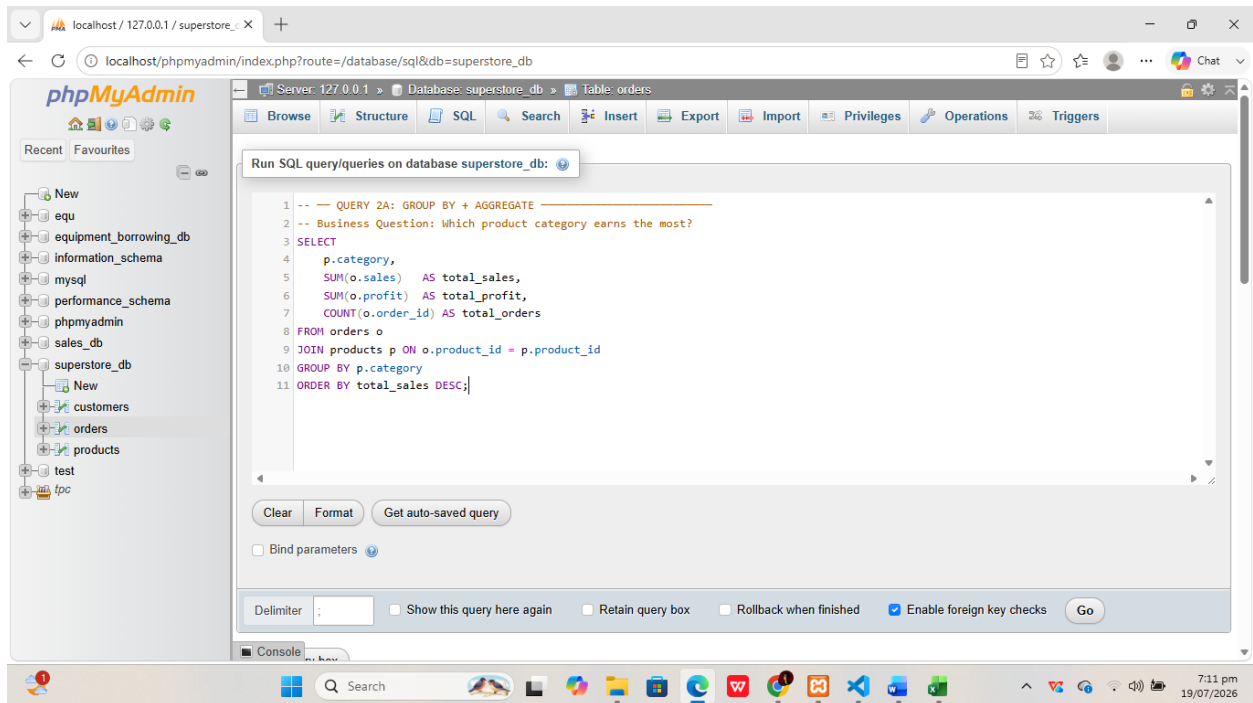
2. Aggregate Analysis (2 Queries)

Must include **GROUP BY** and Aggregate Functions (**COUNT**, **SUM**, **AVG**, etc.).

QUERY 2A: GROUP BY + AGGREGATE

-- Business Question: Which product category earns the most?

- **Screenshot of SQL Code**



- **Result Screenshot**

Server: 127.0.0.1 > Database: superstore_db > Table: p

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 3 (4 total, Query took 0.0172 seconds.)

QUERY 2A: GROUP BY + AGGREGATE -- Business Question: Which product category earns the most? `SELECT p.category, SUM(o.sales) AS total_sales, SUM(o.profit) AS total_profit, COUNT(o.order_id) AS total_orders FROM orders o JOIN products p ON o.product_id = p.product_id GROUP BY p.category ORDER BY total_sales DESC;`

Number of rows: 25 Filter rows: Search this table

category	total_sales	total_profit	total_orders
Technology	63800.64	6775.78	132
Furniture	55643.67	-2865.62	144
Office Supplies	50372.46	8619.42	402
category	0.00	0.00	1

Number of rows: 25 Filter rows: Search this table

Query results operations: Print, Copy to clipboard, Export, Display chart, Create view

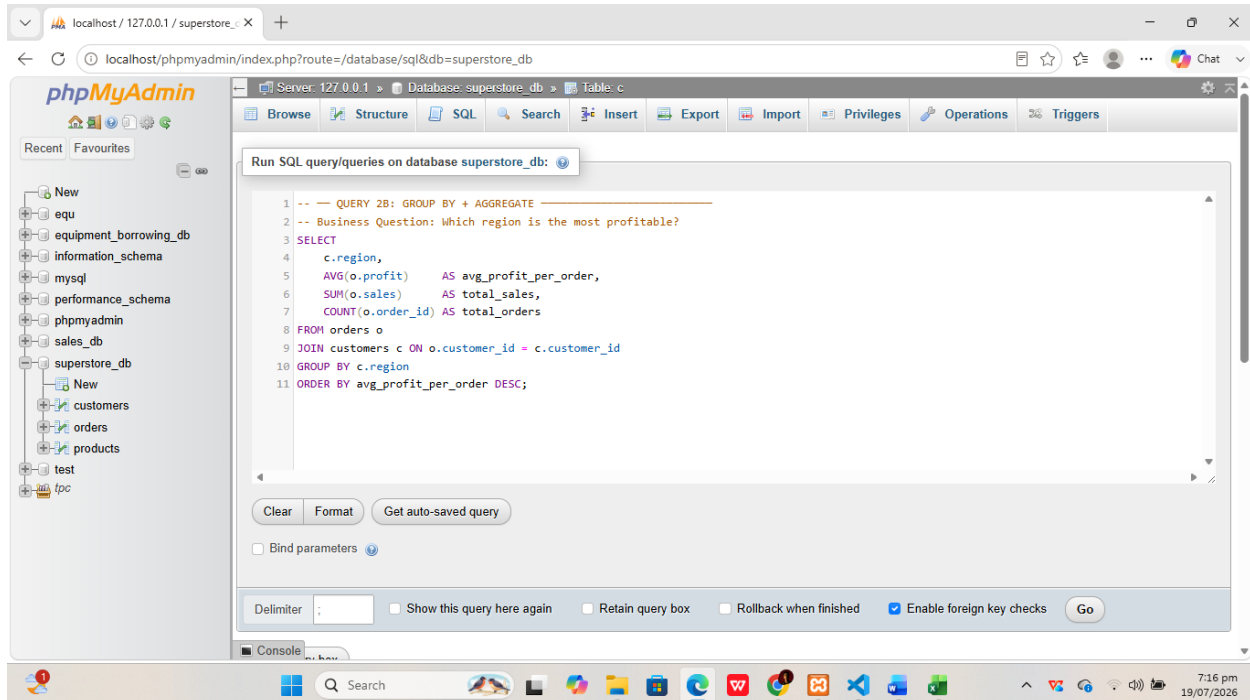
- **Business Interpretation:**

Query 2A: Calculates total sales revenue and total profit per product category. Results reveal which of the three main categories — Furniture, Office Supplies, or Technology — generates the most revenue and profit, informing product mix and resource allocation decisions.

QUERY 2B: GROUP BY + AGGREGATE

-- Business Question: Which region is the most profitable?

- **Screenshot of SQL Code**



- **Result Screenshot**

Server: 127.0.0.1 > Database: superstore_db > Table: c

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 4 (5 total, Query took 0.0021 seconds.)

QUERY 2B: GROUP BY + AGGREGATE -- Business Question: Which region is the most profitable? `SELECT c.region, AVG(o.profit) AS avg_profit_per_order, SUM(o.sales) AS total_sales, COUNT(o.order_id) AS total_orders FROM orders o JOIN customers c ON o.customer_id = c.customer_id GROUP BY c.region ORDER BY avg_profit_per_order DESC;`

Number of rows: 25 Filter rows: Search this table

region	avg_profit_per_order	total_sales	total_orders
South	48.494079	21506.05	76
West	26.475659	47485.38	205
East	10.535860	44073.94	215
Central	6.325879	56751.40	182
region	0.000000	0.00	1

Number of rows: 25 Filter rows: Search this table

- **Business Interpretation:**

Query 2B: Computes average profit per order and total order count by geographic region. A region with many orders but low average profit may indicate heavy discounting or inefficiencies, while a high-profit region signals a target for increased investment.

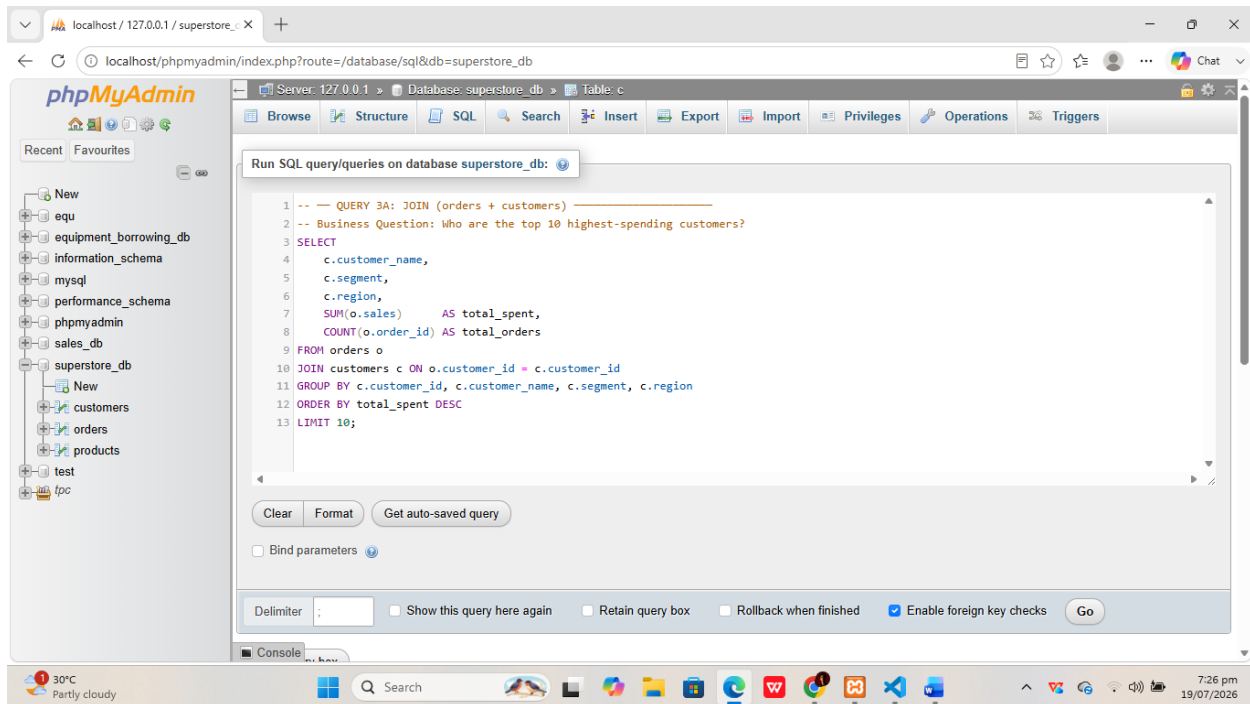
3. Multi-Table Joins (2 Queries)

Must connect two or more tables.

QUERY 3A: JOIN (orders + customers)

--Business Question: Who are the top 10 highest-spending customers?

- **Screenshot of SQL Code**



- **Result Screenshot**

The screenshot shows the phpMyAdmin interface for a database named 'superstore_db'. A SQL query has been executed, and the results are displayed in a table. The query is a JOIN between the 'orders' and 'customers' tables to find the top 10 highest-spending customers. The results table has columns: customer_name, segment, region, total_spent, and total_orders. The data is sorted by total_spent in descending order.

customer_name	segment	region	total_spent	total_orders
Becky Martin	Consumer	Central	10539.90	7
Christopher Martinez	Consumer	South	6412.77	4
Sean Braxton	Corporate	Central	5579.94	2
Nathan Mautz	Home Office	East	4807.37	5
Alan Dominguez	Home Office	Central	4789.94	5
Adam Bellavance	Home Office	East	4389.57	4
Keith Herrera	Consumer	West	3955.51	9
Tracy Blumstein	Consumer	East	3761.14	11
Dianna Wilson	Home Office	Central	3745.63	5
Brosina Hoffman	Consumer	West	3714.29	7

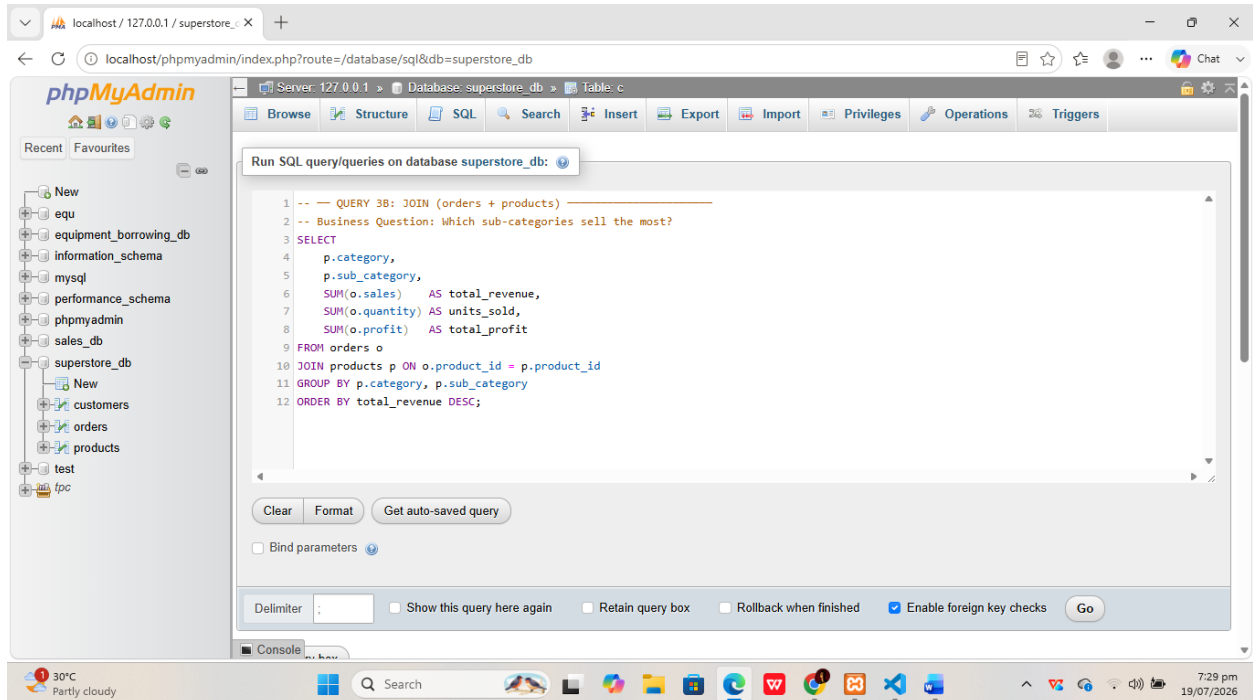
- **Business Interpretation**

Query 3A: Joins orders and customers tables to identify the top 10 customers by total spending. Knowing which customers generate the most revenue enables the business to prioritize relationship management with high-value clients and design targeted loyalty programs.

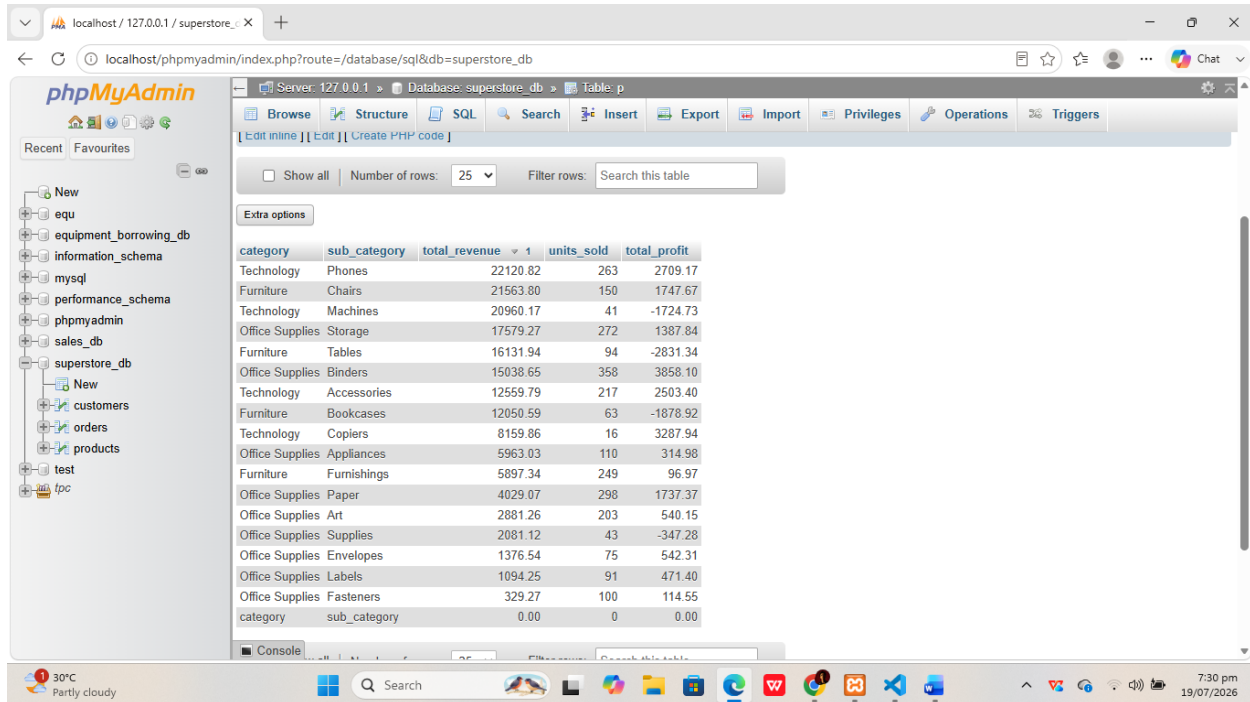
QUERY 3B: JOIN (orders + products)

-- Business Question: Which sub-categories sell the most?

- **Screenshot of SQL Code**



- **Result Screenshot**



The screenshot shows the phpMyAdmin interface for a database named 'superstore_db'. The table 'p' is selected, and its data is displayed in a table format. The table has 5 columns: 'category', 'sub_category', 'total_revenue', 'units_sold', and 'total_profit'. The data is sorted by 'total_revenue' in descending order. The table contains 25 rows of data.

category	sub_category	total_revenue	units_sold	total_profit
Technology	Phones	22120.82	263	2709.17
Furniture	Chairs	21563.80	150	1747.67
Technology	Machines	20960.17	41	-1724.73
Office Supplies	Storage	17579.27	272	1387.84
Furniture	Tables	16131.94	94	-2831.34
Office Supplies	Binders	15038.65	358	3858.10
Technology	Accessories	12559.79	217	2503.40
Furniture	Bookcases	12050.59	63	-1878.92
Technology	Copiers	8159.86	16	3287.94
Office Supplies	Appliances	5963.03	110	314.98
Furniture	Furnishings	5897.34	249	96.97
Office Supplies	Paper	4029.07	298	1737.37
Office Supplies	Art	2881.26	203	540.15
Office Supplies	Supplies	2081.12	43	-347.28
Office Supplies	Envelopes	1376.54	75	542.31
Office Supplies	Labels	1094.25	91	471.40
Office Supplies	Fasteners	329.27	100	114.55
category	sub_category	0.00	0	0.00

- **Business Interpretation**

Query 2B: Joins orders and products tables to rank all sub-categories by total revenue and units sold, providing granular insight into which specific sub-categories like Phones, Chairs, or Binders are the strongest performers.

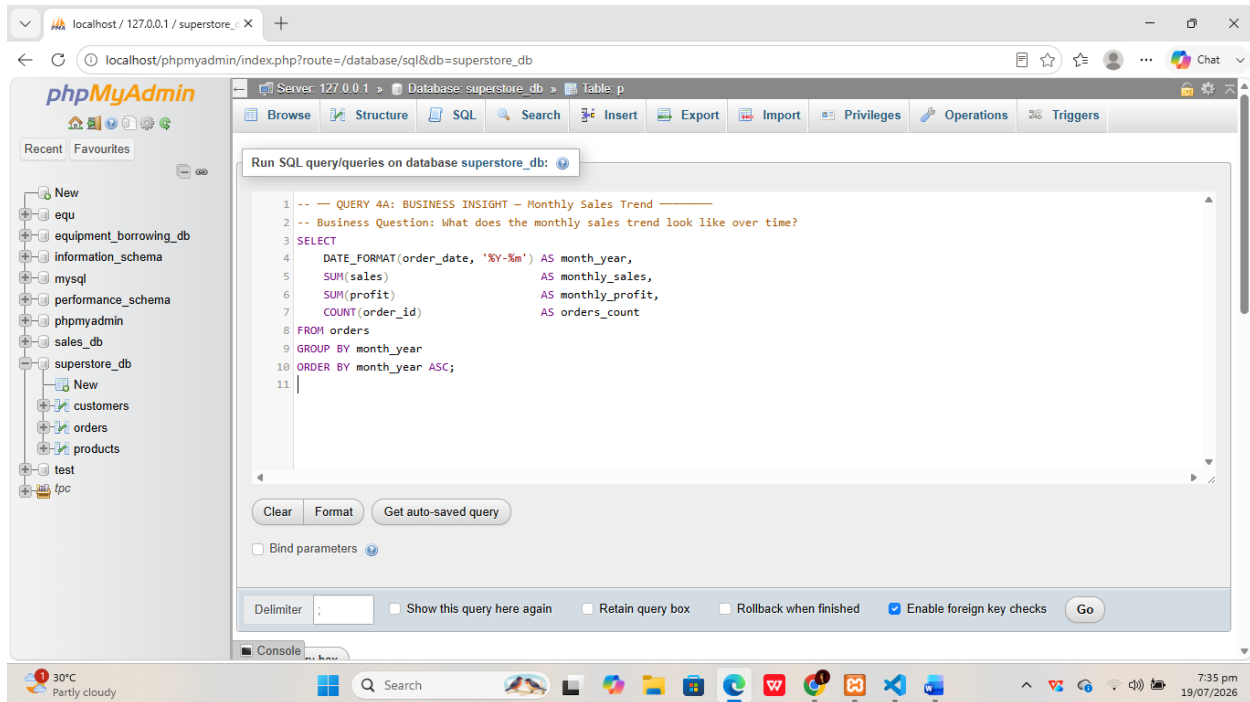
4. Business-Relevant Insights (2 Queries)

Queries specifically designed to answer your key business questions listed in Task 1.1.

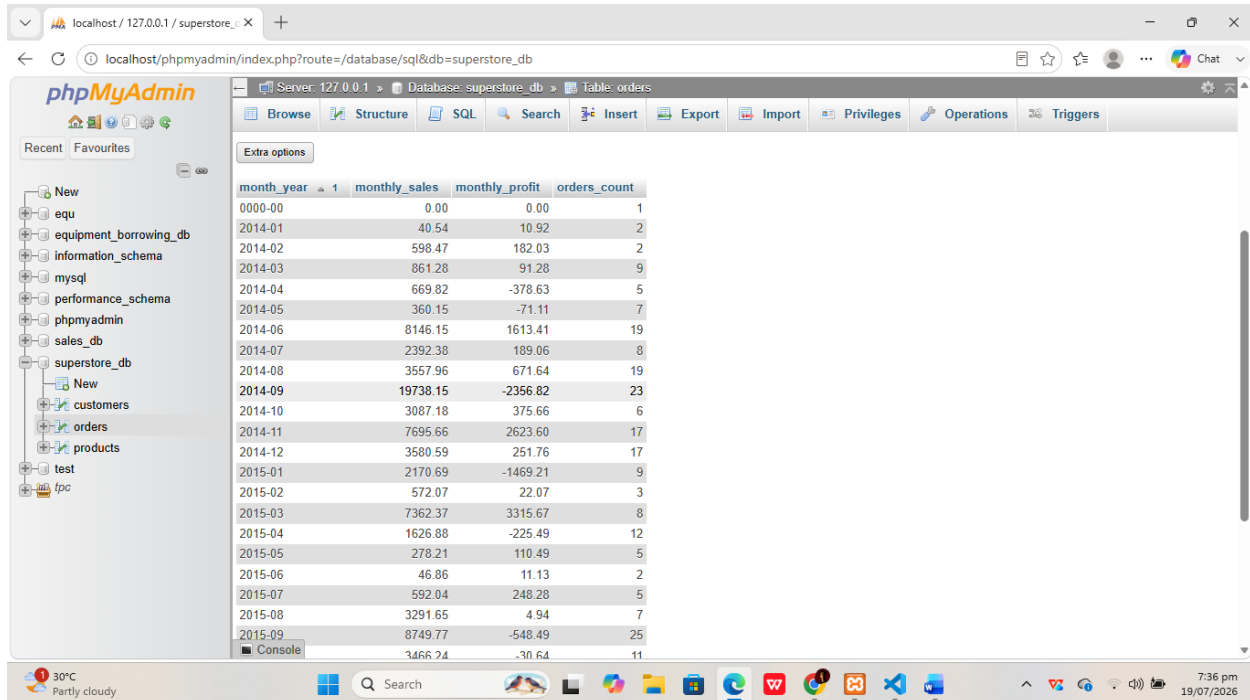
QUERY 4A: BUSINESS INSIGHT — Monthly Sales Trend (Question 3)

-- Business Question: What does the monthly sales trend look like over time?

- **Screenshot of SQL Code**



- **Result Screenshot**



month_year	monthly_sales	monthly_profit	orders_count
0000-00	0.00	0.00	1
2014-01	40.54	10.92	2
2014-02	598.47	182.03	2
2014-03	861.28	91.28	9
2014-04	669.82	-378.63	5
2014-05	360.15	-71.11	7
2014-06	8146.15	1613.41	19
2014-07	2392.38	189.06	8
2014-08	3557.96	671.64	19
2014-09	19738.15	-2356.82	23
2014-10	3087.18	375.66	6
2014-11	7695.66	2623.60	17
2014-12	3580.59	251.76	17
2015-01	2170.69	-1469.21	9
2015-02	572.07	22.07	3
2015-03	7362.37	3315.67	8
2015-04	1626.88	-225.49	12
2015-05	278.21	110.49	5
2015-06	46.86	11.13	2
2015-07	592.04	248.28	5
2015-08	3291.65	4.94	7
2015-09	8749.77	-548.49	25
2015-10	3466.24	-30.64	11

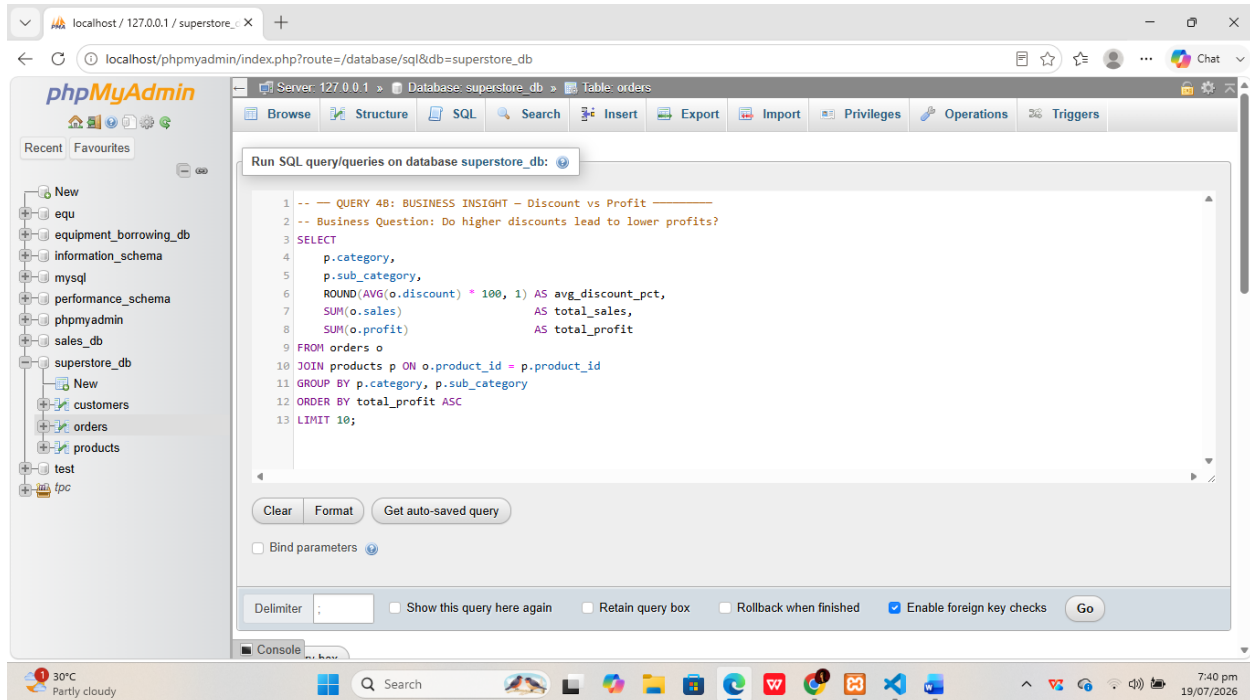
- **Business Interpretation**

Query 4A: Groups total sales and profit by month to reveal peak sales periods such as the November–December holiday season and slower months. This time-series output is essential for demand forecasting and inventory planning for upcoming quarters, directly answering the business question about seasonal trends.

QUERY 4B: BUSINESS INSIGHT — Discount vs Profit (Question 2)

-- Business Question: Do higher discounts lead to lower profits?

- **Screenshot of SQL Code**



- **Result Screenshot**

Server: 127.0.0.1 » Database: superstore_db » Table: p

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 9 (10 total, Query took 0.0030 seconds.)

QUERY 4B: BUSINESS INSIGHT - Discount vs Profit -- Business Question: Do higher discounts lead to lower profits? `SELECT p.category, p.sub_category, ROUND(AVG(o.discount) * 100, 1) AS avg_discount_pct, SUM(o.sales) AS total_sales, SUM(o.profit) AS total_profit FROM orders o JOIN products p ON o.product_id = p.product_id GROUP BY p.category, p.sub_category ORDER BY total_profit ASC LIMIT 10;`

category	sub_category	avg_discount_pct	total_sales	total_profit
Furniture	Tables	31.3	16131.94	-2831.34
Furniture	Bookcases	25.1	12050.59	-1878.92
Technology	Machines	40.0	20960.17	-1724.73
Office Supplies	Supplies	12.3	2081.12	-347.28
category	sub_category	0.0	0.00	0.00
Furniture	Furnishings	17.5	5897.34	96.97
Office Supplies	Fasteners	8.9	329.27	114.55
Office Supplies	Appliances	25.9	5963.03	314.98
Office Supplies	Labels	7.3	1094.25	471.40
Office Supplies	Art	7.9	2881.26	540.15

- **Business Interpretation**

Query 4B: Identifies the 10 least profitable sub-categories along with their average discount rates. Sub-categories with high average discounts and low or negative profit confirm that aggressive discounting is hurting margins, providing actionable data to revise the discounting strategy.

Time Finished – 19/07/2026, 7:43 pm